
Modeling User Preference for Personalized Video Question Answering in Egocentric Video

Ethan Chang Alice Xiao Celinda Zhu
Massachusetts Institute of Technology

Abstract

Egocentric video assistants must often infer not only what a person is doing, but why they are doing it. In household settings, the same local motion can have different explanations for different users because people vary in tool choice, ordering of subtasks, cleanup habits, tempo, and workspace organization. We study whether such user preference information can improve fine-grained why-recognition in egocentric video question answering. The main challenge is that preference is not directly annotated in HD-EPIC, and naively injecting a user profile can distract a vision-language model from visual evidence. We formulate preference-conditioned VQA in which a user-level profile z_u is inferred from long-context history h_u and added to a multimodal model answering a current video question. We compare LLM-derived numbered ratings, LLM paragraph summaries, LLM past-behavior memories, semi-manual hand-labeled profiles, randomized-profile controls, and an action-conditioned gated prompting strategy. On 500 HD-EPIC fine-grained why-recognition questions, a Qwen3-VL-4B baseline achieves 55.0% accuracy. LLM-generated preference prompts generally do not help: numbered ratings, paragraph summaries, and past-behavior memory achieve 51.6%, 50.4%, and 54.2%, while their randomized counterparts achieve 51.6%, 51.6%, and 54.0%. This near-equivalence suggests that the LLM profiles are not strongly user-specific. We also test explicit reasoning-output prompts: baseline answer reasoning reaches 53.4%, and a separate preference-reasoning prompt falls to 47.6%, suggesting that verbalized reasoning can amplify noisy profile evidence rather than improve grounding. Hand-labeled profiles improve to 56.6%, while randomized hand-labeled profiles drop to 53.4%. Hand-labeled profiles with action-conditioned gating achieve the best result, 58.6%, and the randomized gated control is 53.4%. Our analysis shows that preference is useful only when it is concrete, participant-specific, and selectively applied; otherwise, it can cause plausible but incorrect regressions.

1 Introduction

Future household robots and wearable assistants will operate in environments where there is no single canonical way to complete a task. Even in the same kitchen, one person may clean as they go, another may stage all ingredients before cooking, and another may repeatedly use the same tool or measurement routine. These differences are not noise from the user’s perspective. They are persistent behavioral preferences that help explain why an action is being performed. For example, picking up a towel may mean cleaning the counter for one user, drying hands for another, or making room for another object depending on the visible context and the user’s routine. A model that treats all users as exchangeable may answer with generic kitchen commonsense rather than the specific reason for the action.

This problem is especially important for egocentric video question answering (VQA), where the camera observes hands, objects, and short-term motions but often misses the broader behavioral

context. In fine-grained why-recognition, the correct answer is frequently not the most generic reason for an action, but a particular local goal: freeing a hand, making room for a bowl, preventing an ingredient from sticking, recording a measurement, or continuing a personal workflow. Existing egocentric video models are increasingly strong at recognizing actions and objects, but they usually condition on the current clip and question alone. This makes it difficult to distinguish task intent from user-specific habits, especially when the answer choices are semantically close.

The core research challenge is that user preference is both useful and dangerous. It is useful because a user’s long-term history can reveal repeated patterns that are not visible in a single short clip. It is dangerous because preference descriptions are approximate: they can be coarse, noisy, irrelevant to the current action, or overly generic. A model that over-trusts preference may ignore the video and regress from a correct visually grounded answer to an incorrect profile-driven answer. Therefore, our goal is not only to ask whether preference helps, but also when and how it should be used.

In this paper, we study preference-conditioned VQA for fine-grained why-recognition in egocentric cooking videos. We make three contributions. First, we formalize user preference as a latent user-level variable z_u inferred from long-context history h_u and used to condition a VQA model on a current clip and question. Second, we compare several practical preference construction methods: LLM-generated numbered ratings, LLM paragraph summaries, LLM-extracted past-behavior memory, semi-manual hand-labeled profiles, randomized counterparts for these profile families, and hand-labeled profiles with action-conditioned gating. Third, we analyze the results through participant-level accuracy, action-verb accuracy, fixed/regressed examples, and true-versus-random profile controls. Our main finding is that preference is not a plug-and-play improvement: naive LLM profiles hurt and behave similarly to their randomized controls, hand-labeled profiles help modestly, and gated hand-labeled profiles work best because they reduce cases where preference overwhelms visual evidence.

2 Related Work

Egocentric video understanding. Egocentric video benchmarks have moved from recognizing short actions toward understanding long-horizon activities, object interactions, and future-consistent behavior. EPIC-KITCHENS-100 introduced a large-scale benchmark for unscripted kitchen activities and action anticipation [1]. Ego4D expanded egocentric video understanding across a wide range of daily activities and tasks, including memory, forecasting, and social interaction [2]. HD-EPIC further emphasizes high-density egocentric annotations and fine-grained reasoning in cooking environments [3]. Video-language pretraining approaches such as EgoVLP and LaViLa learn representations that align egocentric video with natural language and improve downstream action and narration tasks [4, 5]. These models capture strong visual and linguistic priors, but they typically predict a generic answer from the current clip rather than explicitly modeling persistent user-level variation.

Vision-language models for VQA. Large vision-language models have become powerful zero-shot and few-shot reasoners by combining visual encoders with pretrained language models. CLIP demonstrated scalable image-text alignment [6], while Flamingo, BLIP-2, LLaVA, MiniGPT-4, Video-ChatGPT, and Video-LLaVA explored instruction-following and multimodal dialogue over images and videos [7–12]. These models can answer multiple-choice video questions through prompting without task-specific training. However, they are also sensitive to prompt wording and can be biased by irrelevant textual context. Our work uses a prompt-only VLM setting to test whether adding user preference text improves why-reasoning, and our results show that unfiltered preference text can degrade performance.

Personalization and user modeling. Personalization is well studied in recommender systems, but it is less developed in egocentric vision. Prior work has explored personalized video summarization and personalized object detection for assistive technologies [13, 14]. Federated and personalized learning approaches also adapt models to user-specific data distributions [15]. Our project differs from standard personalization by treating the user signal not only as a distribution shift, but as an explanatory variable for why a user performs an action. The profile is meant to encode behavioral style, such as whether a participant cleans immediately, stages tools, uses a scale frequently, or prefers certain workflows.

Intent inference, theory of mind, and preference conditioning. Why-recognition is related to intent inference and theory of mind because the model must infer hidden goals and beliefs from visible behavior. Watch-and-Help studies intent inference for collaborative assistance [16], and EgoToM evaluates theory-of-mind reasoning from egocentric videos, including goals, beliefs, and next actions [17]. Latent-goal methods model unobserved goals to anticipate future actions [18]. Our work is complementary: instead of only modeling the goal of the current task, we decouple goal from preference. Two users may share the same goal, such as preparing a meal, but differ in how they execute it. Methodologically, our gated prompting is inspired by feature-wise conditioning and gated multimodal fusion, which selectively modulate representations or inputs based on context [19, 20]. We adapt this idea to a prompt-based VLM by gating the relevance of a textual user profile according to the action type.

3 Problem Statement

Let $u \in \mathcal{U}$ denote a participant. For each VQA example i , we observe an egocentric video clip v_i , a natural-language question q_i , and a set of five answer choices $C_i = \{c_{i1}, \dots, c_{i5}\}$. The label is $y_i \in \{1, \dots, 5\}$, indicating the correct answer choice. We write the current VQA input as

$$x_i = (v_i, q_i, C_i). \quad (1)$$

The standard baseline predicts

$$\hat{y}_i = \arg \max_{k \in \{1, \dots, 5\}} p_\theta(c_{ik} \mid x_i), \quad (2)$$

where p_θ is a pretrained multimodal VQA model.

To model user-specific variation, we introduce a user preference variable $z_u \in \mathcal{Z}$. The preference is inferred from a participant history h_u , which consists of long-context egocentric video sampled from the same user but outside the current short VQA clip. In the most general form,

$$z_u = g_\phi(h_u), \quad \hat{y}_i = \arg \max_k p_\theta(c_{ik} \mid x_i, z_u). \quad (3)$$

In a trained setting, one could optimize

$$\mathcal{L}(\theta, \phi) = - \sum_i \log p_\theta(y_i \mid x_i, g_\phi(h_{u_i})). \quad (4)$$

In this project, we do not fine-tune the VLM. Instead, we instantiate g_ϕ through LLM-generated or human-curated profiles and inject z_u through prompts. We also define an action-conditioned gate $\gamma(a_i)$, where a_i is the action verb extracted from the question. The gated model predicts

$$\hat{y}_i = \arg \max_k p_\theta(c_{ik} \mid x_i, \gamma(a_i) \odot z_{u_i}), \quad (5)$$

where $\gamma(a_i)$ is implemented as prompt instructions that make the model visual-heavy, preference-heavy, or balanced.

4 Proposed Method

4.1 Overview

Our method keeps the VQA backbone fixed and varies only the preference information provided in the prompt. This lets us isolate whether a user-level profile improves fine-grained why-recognition without introducing additional training instability. Figure 1 summarizes the pipeline. A video history for each participant is converted into a preference profile. Each video history is created by selecting a combination of videos that summed over 60 minutes. At inference time, the 32 seconds related window in the current video clip and multiple choice question are paired with the participant profile. In the gated variant, the action verb determines whether the prompt should emphasize visual evidence, user preference, or both.

4.2 Preference construction

Because HD-EPIC does not provide explicit preference annotations, we construct participant-level profiles from long-context history. We explored five profile formats.

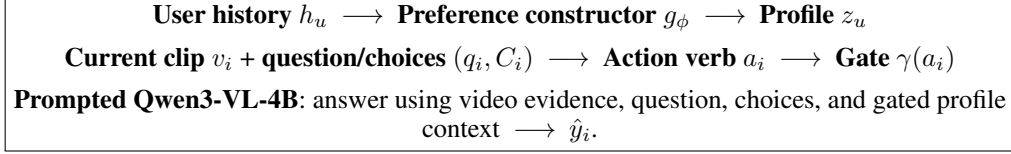


Figure 1: Preference-conditioned VQA pipeline. The user profile is derived from long-context participant history, while the action gate controls how strongly that profile should influence the current answer.

LLM numbered ranking. We used the selected video history of each participant and used Gemini to assign numerical ratings to eight behavioral traits. These traits are tempo, precision, tool preference, cleanliness, workflow organization, ingredient staging, measurement style, and heat or timing management. Each dimension is represented as a value on a 1 to 5 scale, producing a structured vector $z_u \in \{1, 2, 3, 4, 5\}^8$. The prompt includes these ratings and short explanations for each trait.

LLM paragraph summary. Instead of structured ratings, Gemini produces a natural language summary of the participant’s cooking style. This format is more expressive, but it may also be vague or overly general. The prompt includes these paragraphs.

LLM past-behavior memory. We extracted a set of 6 to 8 concrete past behaviors from the user’s videos, such as repeated measurement actions, cleanup routines, or tool choices. The 5 behaviors with the most confidence was selected and added to the prompt. The hypothesis is that concrete memories may be more useful than abstract traits because why-questions often require local procedural reasoning.

Semi-manual hand-labeled profile. After observing that LLM-generated profiles were noisy, we created more detailed hand-labeled profiles. We watched the selected video history for each participant. Then, we summarized the LLM labels, retained traits that appeared relevant, and added concrete details. These profiles are still text prompts, but they are more grounded in observed participant behavior and less generic than the automatically generated summaries.

Counterfactual Preference Editing. To validate whether the model meaningfully uses user preference information, we evaluate a randomized preference label version for the numbered, paragraph, past-behavior, hand-labeled, and gated hand-labeled settings. The input video and question x are held fixed, and the prompt format remains unchanged, while the preference profile is replaced with a profile from the wrong participant. This allows us to compare the resulting predictions $p(y | x, z)$ and $p(y | x, z')$, where z denotes the real participant profile and z' denotes a counterfactual randomized profile. If accuracy remains similar after randomization, the model may be ignoring the preference signal or benefiting only from the presence of additional generic context. In contrast, a drop in accuracy relative to the real profile condition suggests that the original profile contains user specific behavioral information that helps resolve ambiguity in the why-recognition task. More broadly, consistent and interpretable changes under counterfactual profiles would indicate that the model is using preference as an explanatory factor rather than treating it as irrelevant prompt text.

4.3 Prompt-based preference-conditioned VQA

For each example, we prompt Qwen3-VL-4B with 32 sampled frames (one for each second in the 32 second context window), the question, and five answer choices. The baseline prompt contains only the video and VQA question. The preference conditioned prompt additionally includes a participant profile:

$$P_i = \text{Prompt}(v_i, q_i, C_i, z_{u_i}). \quad (6)$$

The model is instructed to choose exactly one answer option. We parse the predicted option and compute exact-match accuracy against the ground-truth choice.

This prompt only design has two advantages. First, it is lightweight and does not require training a new video model. Second, it directly tests whether current VLMs can use natural-language user

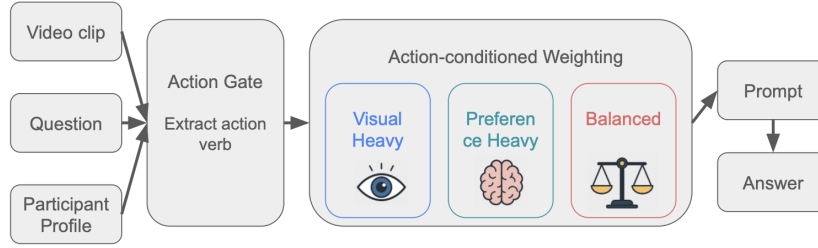


Figure 2: Gated preference weighting pipeline

histories as side information. The limitation is that the model has no learned mechanism for deciding profile relevance.

Reasoning prompt variants. On top of the different representation of preference, we also tried adding reasoning in the prompt. The prompt first asks for a preference specific rationale about which parts of the user history are relevant, and then asks for the final answer. These variants test whether forcing the model to verbalize its intermediate reasoning helps it use profile context. Accuracy is still computed only from the final selected answer.

4.4 Action conditioned gated preference weighting

The final method introduces a gate for the weight of the preference in the prompt. Figure 2 shows the pipeline. It was inspired by the gated fusion method we learned in lectures. The key idea is that not all actions should use preference equally. Some actions, such as transferring an object or moving a pan, often have strong visual evidence and should be answered mostly from the current clip. Other actions, such as tapping a scale or placing a spoon, may be more related to a participant’s repeated habits and should be answered with the help from preference profile. We therefore extract the action verb and map it to one of three modes:

- **Visual heavy:** prioritize current visual evidence, use profile only if it directly resolves ambiguity.
- **Preference heavy:** use participant profile to choose between plausible visually consistent explanations.
- **Balanced:** combine video evidence and preference evidence without strongly privileging either.

In implementation, the gate is a prompt level instruction. For example, a visual heavy prompt explicitly tells the model that the current clip should override the profile if they conflict. This design tests whether relevance control can reduce the regressions caused by unconditional preference profile injection.

5 Experimental Methodology

Dataset. We evaluate on the HD-EPIC VQA subset `fine_grained_why_recognition`. The benchmark contains 500 multiple-choice questions, each asking why a person performed a specific action in an egocentric cooking video. The data includes 9 participants, with long egocentric cooking videos for each participant. Each example has an action timestamp, an egocentric video clip, a natural-language why-question, five answer choices, and a ground-truth answer. The dataset does not provide explicit preference annotations, which motivates our profile construction procedure.

Video and model input. For each VQA example, we sample 32 frames at 1 FPS: 16 seconds before and 16 seconds after the provided timestamp. This gives the VLM short-term context before and after the queried action. We use Qwen3-VL-4B as the backbone model under a multiple-choice prompting setup. All compared methods use the same video frames and answer choices; only the profile context and gating instructions differ.

Evaluation setting. Our experiments are evaluation-only prompt ablations rather than supervised training experiments. We do not split the 500 questions into training and test sets because no model parameters are learned. Participant profiles are constructed at the participant level from long-context video history and then reused across that participant’s VQA examples. We report exact-match accuracy, participant-level accuracy, action-verb accuracy, and fixed/regressed counts relative to the baseline.

Compared methods. We compare thirteen settings: (1) no-preference baseline; (2) LLM numbered ratings; (3) LLM paragraph summary; (4) LLM past-behavior memory; (5–7) randomized versions of the three LLM profile formats; (8) hand-labeled profile; (9) randomized hand-labeled profile; (10) hand-labeled profile with action-conditioned gating; (11) randomized hand-labeled profile with the same gate; (12) baseline reasoning; and (13) preference reasoning. The randomized conditions are counterfactual controls for profile identity, while the reasoning settings test whether explicit generated rationales improve answer selection.

6 Results and Discussion

6.1 Overall accuracy

Table 1 summarizes the main results. The baseline achieves 55.0% accuracy, corresponding to 275 correct answers out of 500. The three LLM-generated preference formats do not improve the baseline. Numbered ratings and paragraph summaries are substantially below the baseline, and past-behavior memory is only slightly below it.

The randomized LLM controls are especially informative. Randomized numbered ratings match the true numbered ratings exactly at 51.6%; randomized paragraph summaries reach 51.6%, above the true paragraph result of 50.4%; and randomized past-behavior memory reaches 54.0%, only one correct answer below the true memory result of 54.2%. This pattern suggests that the LLM-generated profiles add prompt context but do not reliably encode information that is specific to the correct participant.

Explicit reasoning prompts also do not improve performance. Baseline reasoning, which asks for a final-answer rationale before the choice, reaches 53.4% (267/500). Preference reasoning, which adds a separate preference-specific reasoning step, falls to 47.6% (238/500). This suggests that asking the model to explain how preference applies can make it over-commit to noisy or weakly relevant profile evidence.

Hand-labeled profiles are more useful. The real hand-labeled profile reaches 56.6% (283/500), while the randomized hand-labeled profile drops to 53.4% (267/500). The best method is the real hand-labeled profile plus action-conditioned gating, which reaches 58.6% (293/500), an 18-example improvement over the baseline and a 10-example improvement over hand-labeled profiles without gating. Its randomized gated control is again 53.4% (267/500). This control is important: it suggests that the gain is not simply due to adding profile-like text or making the prompt longer. The identity and relevance of the profile matter.

The improvement from 55.0% to 58.6% is modest, so it should not be interpreted as a solved problem. However, it is meaningful as a directional result because the gain comes from a simple prompt-level intervention without fine-tuning. The paired fixed/regressed analysis in Table 2 explains where the gain comes from. The real hand-labeled profile fixes 47 baseline mistakes but introduces 39 new mistakes. Gating increases the number of fixed examples to 52 and reduces regressions to 34. In contrast, baseline reasoning fixes 51 but regresses 59, and preference reasoning fixes 40 but regresses 77. In other words, the gate does not only add more preference signal; it also prevents some harmful overuse of the profile, while explicit reasoning can increase regressions.

6.2 Participant-level performance

Table 3 shows that the effect of preference conditioning varies substantially by participant. The gated hand-labeled method improves over the baseline for P02, P03, P04, P05, P06, P07, and P08; it is unchanged for P09 and worse for P01. The largest gains occur for P04 (+9.1 percentage points), P05 (+7.7), P07 (+6.6), P03 (+6.4), and P02 (+6.2). These results support the hypothesis that some participants exhibit stable behavioral regularities that help answer why-questions.

Table 1: Overall accuracy on 500 HD-EPIC fine-grained why-recognition questions. All rows are computed from the CSV logs. “Shuffled” means the profile came from a different participant; the reasoning rows use the true past-behavior memory context but change the requested rationale format.

Method	Profile identity	Gate	Correct	Accuracy
Baseline	none	no	275	55.0%
LLM numbered ranking	true	no	258	51.6%
LLM paragraph summary	true	no	252	50.4%
LLM past-behavior memory	true	no	271	54.2%
Baseline reasoning prompt	true	no	267	53.4%
Preference reasoning prompt	true	no	238	47.6%
Random LLM numbered ranking	shuffled	no	258	51.6%
Random LLM paragraph summary	shuffled	no	258	51.6%
Random LLM past-behavior memory	shuffled	no	270	54.0%
Hand-labeled profile	true	no	283	56.6%
Random hand-labeled profile	shuffled	no	267	53.4%
Hand-labeled profile + gate	true	yes	293	58.6%
Random hand-labeled profile + gate	shuffled	yes	267	53.4%

Table 2: Fixed and regressed examples relative to the baseline. A fixed example is wrong for the baseline but correct for the method; a regressed example is correct for the baseline but wrong for the method. Net is fixed minus regressed.

Method	Fixed	Regressed	Net	Changed pred.
LLM numbered ranking	15	32	-17	86
LLM paragraph summary	17	40	-23	92
LLM past-behavior memory	36	40	-4	120
Baseline reasoning prompt	51	59	-8	164
Preference reasoning prompt	40	77	-37	181
Random LLM numbered ranking	14	31	-17	85
Random LLM paragraph summary	20	37	-17	91
Random LLM past-behavior memory	33	38	-5	108
Hand-labeled profile	47	39	+8	136
Random hand-labeled profile	38	46	-8	122
Hand-labeled profile + gate	52	34	+18	140
Random hand-labeled profile + gate	35	43	-8	121

The random controls show why profile identity still matters. For example, P04 improves from 72.7% to 81.8% with the real gated profile, but falls to 65.9% with the randomized gated profile. P08 is 56.3% with the real gated profile and 49.4% with the randomized gated profile. At the same time, preference is not uniformly helpful: P01 drops from 55.9% to 50.0% under the real gated profile, and P09 remains flat at 49.1%. These cases suggest that some profiles may be too broad, not sufficiently aligned with the queried clips, or unable to resolve the relevant ambiguity. They also show why a learned relevance mechanism would likely be preferable to a hand-designed verb gate.

6.3 Action-verb breakdown

Table 4 groups examples by the first verb in the action phrase. The gated method helps most on actions where the local motion is visually simple but the reason is ambiguous. For example, accuracy on *tap* increases from 51.1% to 64.4%, on *stir* from 33.3% to 53.3%, and on *place* from 55.6% to 63.9%. These are exactly the kinds of actions where multiple explanations are plausible: tapping may zero a scale, update a measurement, or check a device; stirring may mix, prevent burning, incorporate a new ingredient, or change texture; placing an object may clear workspace, stage an ingredient, avoid mess, or prepare for measurement.

The method does not improve all verbs. For *transfer*, accuracy drops slightly from 44.4% to 42.6%, and for *move*, *rotate*, *open*, and *turn*, performance is roughly unchanged. This pattern supports the motivation for gating: some verbs are mostly determined by visual evidence, while others benefit from user context. However, the current gate is still too simple because it uses only the verb and not the complete scene state, object identity, or answer choices.

Table 3: Participant-level accuracy for the baseline and the hand-labeled profile variants.

Participant	#Q	Baseline	Hand	Rand hand	Hand + gate	Rand gate
P01	34	55.9%	47.1%	52.9%	50.0%	44.1%
P02	48	52.1%	58.3%	50.0%	58.3%	47.9%
P03	78	61.5%	67.9%	57.7%	67.9%	67.9%
P04	44	72.7%	79.5%	70.5%	81.8%	65.9%
P05	52	34.6%	40.4%	40.4%	42.3%	40.4%
P06	13	69.2%	84.6%	69.2%	76.9%	76.9%
P07	30	66.7%	60.0%	60.0%	73.3%	63.3%
P08	87	55.2%	51.7%	54.0%	56.3%	49.4%
P09	114	49.1%	49.1%	47.4%	49.1%	47.4%

Table 4: Action-verb accuracy for verbs with at least 10 questions. “Rand hand” is the randomized hand-labeled profile without gating; “Rand gate” is the randomized hand-labeled profile with the action gate.

Verb	#Q	Base	Hand	Rand hand	Hand + gate	Rand gate
pick	62	56.5%	61.3%	58.1%	62.9%	59.7%
transfer	54	44.4%	40.7%	38.9%	42.6%	40.7%
tap	45	51.1%	62.2%	51.1%	64.4%	57.8%
shake	38	57.9%	55.3%	57.9%	60.5%	63.2%
move	37	70.3%	64.9%	64.9%	70.3%	64.9%
place	36	55.6%	63.9%	52.8%	63.9%	47.2%
put	25	52.0%	44.0%	40.0%	52.0%	32.0%
open	23	52.2%	56.5%	52.2%	52.2%	56.5%
turn	21	52.4%	52.4%	47.6%	52.4%	47.6%
rotate	16	62.5%	62.5%	56.2%	62.5%	62.5%
stir	15	33.3%	46.7%	33.3%	53.3%	40.0%
tilt	14	85.7%	92.9%	78.6%	85.7%	85.7%
lift	12	58.3%	66.7%	58.3%	66.7%	58.3%

6.4 Why LLM-generated preference labels failed

The negative LLM-profile results are informative. The numbered ranking representation compresses a user’s behavior into broad dimensions, but why-recognition often depends on highly local distinctions. A rating such as “cleanliness = high” does not tell the model whether a specific towel action is for drying hands, wiping a counter, or moving the towel away. Paragraph summaries are more expressive, but they often contain generic cooking statements and can bias the model toward common explanations. Past-behavior memory is the strongest LLM-generated variant at 54.2%, but still below the baseline, likely because retrieved memories are not always relevant to the current object or action.

The true-versus-random comparisons make this failure mode clearer. For numbered rankings, the real profile beats the random profile on 7 examples, while the random profile beats the real profile on 7 examples. For paragraph summaries, the real profile wins on 23 examples and the random profile wins on 29. For past-behavior memory, the comparison is nearly tied, 22 versus 21. In other words, the LLM-generated profiles do change predictions, but not in a way that reliably tracks the correct participant. They act more like generic prompt priors than personalized evidence. The reasoning ablation points to the same issue: asking for preference reasoning makes the model more verbose about the profile, but accuracy drops sharply because the generated rationale can justify an irrelevant preference instead of correcting it.

A representative failure is the cheese-grater example from the qualitative analysis. The question asks why the participant picked up a cheese grater. The correct answer is that the participant needed to make room for a large plastic bowl. The baseline and LLM profile variants tend to answer with generic cleaning or cheese-related explanations. The hand-labeled profile and gated prompt recover the correct spatial/workflow reason. This example shows that preference must be tied to concrete workflow and local context; otherwise, it encourages plausible but incorrect commonsense.

6.5 How gating prevents preference regressions

Unconditional hand-labeled preference improves some examples but also creates new errors. The clearest example is the tea-towel question. The ground truth is that the participant picked up the

tea towel to dry hands. The baseline correctly predicts “to dry hands,” but the ungated hand-labeled profile predicts a cleaning-related answer, likely because the profile emphasizes cleanup habits. The gated method preserves the baseline answer because the gate instructs the model to prioritize the current video when profile evidence is not directly relevant. This is exactly the desired behavior: preference should break ties between plausible interpretations, not override strong visual evidence.

The randomized hand-labeled controls further support the importance of true user identity. Without gating, the real hand-labeled profile is correct while the randomized hand-labeled profile is wrong on 46 examples, while the reverse occurs on 30 examples. With gating, this asymmetry becomes stronger: the real gated profile is correct while the randomized gated profile is wrong on 52 examples, while the reverse occurs on 26 examples. This suggests that the gated method is not merely benefiting from additional text; it benefits when the text corresponds to the correct participant and is used in the right action contexts. However, the randomized conditions still solve some examples, which means generic profile language and prompt changes can also influence the model. Future work should separate these effects more carefully with learned controls and profile-relevance labels.

6.6 Remaining error modes

Despite the improvement, many errors remain. First, the video sampling is temporally sparse. We use 32 seconds at 1 FPS, which gives broad context but can miss brief hand-object contacts, small state changes, or quick transitions that matter for why-recognition. Second, the answer choices are often semantically close. A model may select an answer that is plausible in the kitchen context but not the annotated fine-grained intent. Third, the preference profiles are static participant-level summaries. A user may behave differently across recipes or task stages, and a single profile may not capture these changes. Fourth, our gate is hand-designed and verb-based. It cannot tell whether a particular *pick* action is visually obvious, preference-driven, or dominated by object state. These limitations explain why the best method improves accuracy but still fails on 207 of 500 questions.

7 Conclusion and Future Directions

We studied whether user preference can improve fine-grained why-recognition in egocentric video VQA. Our results show a nuanced answer. Adding preference text is not automatically helpful: LLM-generated numbered rankings, paragraph summaries, and past-behavior memories all underperform the no-preference baseline, and their randomized counterparts are nearly as strong as the true profiles. Explicit reasoning-output prompts also do not solve the problem: baseline reasoning reaches 53.4%, and preference reasoning falls to 47.6%. This indicates that coarse profiles and verbal rationales alone do not provide reliable participant-specific evidence. More concrete hand-labeled profiles improve the baseline from 55.0% to 56.6%, while randomized hand-labeled profiles drop to 53.4%. Adding an action-conditioned gate improves further to 58.6%, and the randomized gated control remains at 53.4%. The main lesson is that preference should be treated as a conditional source of evidence rather than a universally useful context field.

The project has several limitations. The evaluation set has only 500 questions and 9 participants, so the gains should be interpreted cautiously. The method is prompt-based and does not train the VLM to use user history. The hand-labeled profiles are more accurate than LLM profiles but do not scale well. The gate is a manually designed heuristic based mostly on the action verb, and it cannot reason about object state, uncertainty, or candidate-answer semantics. Finally, exact-match accuracy is strict and does not capture cases where the model chooses a semantically plausible but differently worded explanation.

Future work should move from prompt-level preference injection to learned preference-aware reasoning. One direction is to fine-tune a VLM or lightweight adapter so that user preference is represented as a learned embedding rather than a long textual prompt. Another is to jointly train the answerer and a relevance gate, allowing the model to decide when profile evidence should be used based on the video, action, object state, and answer choices. A third direction is contrastive preference learning: train on true versus randomized participant profiles so the model learns which behavioral patterns are actually informative. Finally, better long-context profile extraction could use denser video sampling, temporal retrieval, and uncertainty estimates, producing profiles that are specific enough to help but cautious enough not to override visual evidence.

References

- [1] Damen, D., Doughty, H., Farinella, G. M., et al. Revisiting EPIC-KITCHENS: 100 hours of egocentric video. *CVPR*, 2021.
- [2] Grauman, K., Westbury, A., Byrne, E., et al. Ego4D: Around the world in 3,000 hours of egocentric video. *CVPR*, 2022.
- [3] HD-EPIC VQA Challenge. HD-EPIC visual question answering benchmark and challenge materials, 2025.
- [4] Lin, K., Li, L., Lin, C.-C., et al. EgoVLP: Egocentric video-language pretraining. *NeurIPS*, 2022.
- [5] Zhao, Y., Misra, I., Krishnan, R. G., and Girdhar, R. Learning video representations from language labels in egocentric videos. *CVPR*, 2023.
- [6] Radford, A., Kim, J. W., Hallacy, C., et al. Learning transferable visual models from natural language supervision. *ICML*, 2021.
- [7] Alayrac, J.-B., Donahue, J., Luc, P., et al. Flamingo: A visual language model for few-shot learning. *NeurIPS*, 2022.
- [8] Li, J., Li, D., Savarese, S., and Hoi, S. BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. *ICML*, 2023.
- [9] Liu, H., Li, C., Wu, Q., and Lee, Y. J. Visual instruction tuning. *NeurIPS*, 2023.
- [10] Zhu, D., Chen, J., Shen, X., Li, X., and Elhoseiny, M. MiniGPT-4: Enhancing vision-language understanding with advanced large language models. *arXiv preprint*, 2023.
- [11] Maaz, M., Rasheed, H., Khan, S., and Khan, F. S. Video-ChatGPT: Towards detailed video understanding via large vision and language models. *arXiv preprint*, 2023.
- [12] Lin, B., Zhu, B., Ye, Y., Ning, M., Jin, P., and Yuan, L. Video-LLaVA: Learning United Visual Representation by Alignment Before Projection. *arXiv preprint*, 2023.
- [13] Sharghi, A., Laurel, J. S., and Gong, B. Query-guided personalized video summarization of egocentric videos. *arXiv preprint*, 2017.
- [14] Bambach, S., Crandall, D. J., and Yu, C. Personalized object detection for egocentric assistive technologies. *WACV*, 2018.
- [15] Smith, V., Chiang, C.-K., Sanjabi, M., and Talwalkar, A. S. Federated multi-task learning. *NeurIPS*, 2017.
- [16] Puig, X., Shu, T., Li, S., et al. Watch-and-Help: A challenge for social perception and human-AI collaboration. *ICLR*, 2021.
- [17] Li, Y., Veerabadran, V., Iuzzolino, M. L., Roads, B. D., Celikyilmaz, A., and Ridgeway, K. EgoToM: Benchmarking theory of mind reasoning from egocentric videos. *arXiv preprint*, 2025.
- [18] Kim, J., et al. Anticipating next actions via latent goal learning. *ICCV*, 2023.
- [19] Perez, E., Strub, F., de Vries, H., Dumoulin, V., and Courville, A. FiLM: Visual reasoning with a general conditioning layer. *AAAI*, 2018.
- [20] Arevalo, J., Solorio, T., Montes-y-Gomez, M., and Gonzalez, F. A. Gated multimodal units for information fusion. *ICLR Workshop*, 2017.